4.8 Slicing a Spherical Loaf

WORKSHEET

The question

STEP I 2001, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q6

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself The sphere of radius a is obtained by rotating which curve about the x-axis? What is $\frac{dy}{dx}$ for that curve?

STAGE 1 The crust formula simplifies

Ask yourself Compute $y\sqrt{1+(y')^2}$ for the semicircle. Something remarkable happens.

STAGE 3 The crusty condition

Ask yourself Compute V using the given formula. A substitution $s = x + a$ might make the algebra cleaner.

Pause and try The ruling requires $V/A < 1$ *after any number of slices have been eaten*, that is, for every t with $0 < t \leq 2a$. What is the largest value of $\frac{t}{2} - \frac{t^2}{6a}$ on this range?

STAGE 4 Beyond the question: is $2\frac{2}{3}$ enough?

Ask yourself The question asks only for a condition the radius *must* satisfy. Is $a < \frac{8}{3}$ also enough? What if the slices are eaten from both ends?

Try this next

1. **The whole loaf.** Show that for the whole uncut loaf, $V/A = \frac{a}{3}$. Why is the condition in the question harder to satisfy than $\frac{a}{3} < 1$?

2. **The tropics.** What fraction of the Earth's surface lies between latitudes 23.4° south and 23.4° north? (Treat the Earth as a sphere. Since area is proportional to height, you only need $\sin 23.4^\circ$.)

3. **Other shapes.** Show that for a loaf in the shape of a paraboloid (rotate $y = \sqrt{x}$ about the x-axis), the crust per slice is *not* constant. For which curves $y(x)$ is $y\sqrt{1+(y')^2}$ constant? (Solve the differential equation.)

4. **Volumes.** Use the volume formula to show that a sphere has two-thirds the volume

of its wrapping cylinder, as Archimedes found.

Further reading

- **Sherman Stein, *Archimedes: What Did He Do Besides Cry Eureka?*** (Mathematical Association of America). Archimedes' mathematics explained for a modern reader.
- **Timothy Gowers, *Mathematics: A Very Short Introduction*** (Oxford University Press). The chapter on geometry introduces the geometry of curved surfaces.
- **John Snyder, *Flattening the Earth*** (University of Chicago Press). Two thousand years of map projections.

The full guided solution, and what the question leads to at university, is in Section 4.8:
https://beyondstep.keithmcnulty.org/ch04/spherical-loaf.html